

MAT31-1

Linear Algebra III

Euclidean spaces, the spectral theorem, and projection

Add one number — the scalar product — and a vector space gains length, angle, and a notion of "closest".

Albert School — 30 hours

Preface

In **Linear Algebra I** (MAT11-3) you learned the structure of vector spaces: subspaces, bases, dimension, linear maps. In **Linear Algebra II** (MAT21-1) you learned to see **inside** a transformation — the determinant detecting invertibility, the eigenvalues and eigenvectors revealing the directions a matrix merely scales, and diagonalization $A = PDP^{-1}$ exposing a map’s intrinsic content once you choose the right basis. Two pieces of geometry, however, were conspicuously absent from both courses: there was no way to speak of the **length** of a vector, and no way to speak of the **angle** between two vectors. A vector space, as you knew it, was pure algebra without a ruler or a protractor.

This course supplies the missing instrument. Everything flows from adding a single structure — the **scalar product** (or **inner product**) — a way of multiplying two vectors to get a number. From that one number length and angle both emerge, and with them a cascade of geometry: **orthogonality** (vectors at right angles), **orthonormal bases** (the cleanest coordinate systems there are), and the **Gram–Schmidt process** that manufactures them. A vector space equipped with a scalar product is called a **Euclidean space**, and it is the natural home of geometry done with the tools of linear algebra.

The course then reaches two summits. The first is the **spectral theorem**: every real **symmetric** matrix is not merely diagonalizable but **orthogonally** diagonalizable, $A = PDP^T$ with P orthogonal — its eigenvectors can always be chosen mutually perpendicular. This is the cleanest, most useful diagonalization theorem in all of linear algebra, and the eigenvalue-shortfall worries of MAT21-1 simply vanish for symmetric matrices. The second summit is **orthogonal projection** and the **best-approximation theorem**: given a vector and a subspace, there is a unique closest point in the subspace, and we can compute it. The distance to that point is the geometric heart of **least-squares regression**.

These are not abstract curiosities. The scalar product is **cosine similarity**, the workhorse of search and recommendation. The spectral theorem is the engine of **principal component analysis**. Positive-definiteness classifies the critical points found by the **Hessian** in optimization. Best approximation is least-squares. You are building the mathematical foundation of data science.

Keep one slogan in view: **the scalar product turns algebra into geometry — and geometry’s first question is always “how close?”**

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1 The scalar product

Vectors in $\mathbb{R}^n$ can be added and scaled, but so far there has been no way to measure one against another. The **scalar product** is the device that does this: it eats two vectors and returns a single number, and from that number all of length and angle will be reconstructed. We begin with the familiar case in the plane, extract the properties that make it useful, and then promote those properties to a **definition** that works far beyond $\mathbb{R}^n$.

1.1 A first example: the dot product in the plane

You have already met one scalar product, perhaps without the name. For two vectors $u = (u_1, u_2)$ and $v = (v_1, v_2)$ in $\mathbb{R}^2$, their **dot product** is

$$u \cdot v = u_1 v_1 + u_2 v_2.$$

Take $u = (3, 0)$ and $v = (3, 4)$. Then $u \cdot v = 9$. Two facts about this number are worth noticing immediately. First, $u \cdot u = 9 = 3^2$ is the **square of the length** of u . Second, when we compute $v \cdot v = 9 + 16 = 25 = 5^2$, we recover the length 5 of v — exactly Pythagoras. The dot product secretly **knows the geometry** of the plane. Our whole task is to isolate why, and to make the same trick work in spaces where we have no picture to fall back on.

1.2 The axioms

What properties of the dot product are doing the work? Three, and they are all the structure we need.

Definition 1 Let E be a real vector space. A **scalar product** (or **inner product**) on E is a map $\langle \cdot, \cdot \rangle : E \times E \rightarrow \mathbb{R}$ that is

1. **symmetric**: $\langle u, v \rangle = \langle v, u \rangle$ for all u, v ;
2. **bilinear**: linear in each argument separately — for all u, v, w and scalars a, b ,

$$\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$$

(and likewise in the second argument, which then follows by symmetry);

3. **positive definite**: $\langle u, u \rangle \geq 0$ for all u , with $\langle u, u \rangle = 0$ **only** when $u = 0$.

A real vector space equipped with a scalar product is a **Euclidean space** (when finite-dimensional).

The three axioms are not arbitrary. **Symmetry** says the order does not matter. **Bilinearity** says the product interacts with linear combinations exactly the way multiplication should — it is the property that lets us compute. **Positive definiteness** is the crucial one: it guarantees that $\langle u, u \rangle$ is a genuine “squared length”, never negative, and zero only for the zero vector. Drop it and lengths could be imaginary or vanish on non-zero vectors.

Common pitfall All three axioms are required, and positive **definiteness** is stronger than mere positivity. A form can satisfy $\langle u, u \rangle \geq 0$ for all u yet vanish on some non-zero vector — for example $\langle u, v \rangle = u_1 v_1$ on $\mathbb{R}^2$ gives $\langle (0, 5), (0, 5) \rangle = 0$ with $(0, 5) \neq 0$. Such a form is **positive semi-definite** but not a scalar product: it cannot tell the vector $(0, 5)$ apart from 0.

1.3 The standard scalar product on $\mathbb{R}^n$

The dot product of the opening example generalizes verbatim to any dimension.

Definition 2 The **standard** (or **canonical**) scalar product on $\mathbb{R}^n$ is

$$\langle u, v \rangle = u^T v = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

That this satisfies the three axioms is a direct check: symmetry is the commutativity of real multiplication; bilinearity is the distributive law; and $\langle u, u \rangle = \sum u_i^2$ is a sum of squares, hence ≥ 0 and zero only when every $u_i = 0$. This is the scalar product unless another is stated, but — and this is the conceptual leap of the chapter — it is **not the only one**.

1.4 Other Euclidean spaces: polynomials and matrices

The power of the axiomatic definition is that it applies to vector spaces whose elements are not lists of numbers at all. Two examples from the OP matter most.

Worked example — an inner product on polynomials. On the space $\mathbb{R}[X]_{\leq n}$ of polynomials of degree at most n , define

$$\langle P, Q \rangle = \int_0^1 P(x)Q(x) dx.$$

This is symmetric (multiplication of functions commutes) and bilinear (the integral is linear). For positive definiteness, $\langle P, P \rangle = \int_0^1 P(x)^2 dx \geq 0$, and a continuous non-negative function with integral zero must be identically zero — so $\langle P, P \rangle = 0$ forces $P = 0$. Thus polynomials acquire **lengths and angles**: we can ask whether two polynomials are “orthogonal”, a question with no meaning until a scalar product is fixed.

Worked example — an inner product on matrices. On the space $M_n(\mathbb{R})$ of square matrices, the **Frobenius** inner product is

$$\langle A, B \rangle = \text{tr}(A^T B) = \sum_{i,j} a_{ij} b_{ij}.$$

The last equality shows it is **literally** the standard scalar product on $\mathbb{R}^{n^2}$ — flatten each matrix into a long list of its entries and take the dot product. So $\langle A, A \rangle = \sum_{i,j} a_{ij}^2$ is the sum of squared entries, manifestly positive definite.

Remark One vector space can carry **many** different scalar products. On $\mathbb{R}^2$, besides the standard one, $\langle u, v \rangle = 2u_1 v_1 + 3u_2 v_2$ is also a scalar product (a **weighted** dot product). Each choice bends the geometry: lengths and angles depend on which scalar product is in force. When we speak of “the angle between two vectors” we always mean **relative to a chosen scalar product**.

Exercises

1. Verify that $\langle u, v \rangle = 2u_1 v_1 + 3u_2 v_2$ satisfies all three axioms on $\mathbb{R}^2$, and explain where positive definiteness would fail if the coefficient 3 were replaced by 0 or by -1 .
2. On $\mathbb{R}[X]_{\leq 2}$ with $\langle P, Q \rangle = \int_0^1 PQ dx$, compute $\langle 1, X \rangle$ and $\langle 1, 1 \rangle$.
3. For the Frobenius product on $M_2(\mathbb{R})$, compute $\langle I, J \rangle$ where $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and compute $\langle I, I \rangle$.

4. Show that $\langle u, v \rangle = u_1 v_1 - u_2 v_2$ on $\mathbb{R}^2$ is symmetric and bilinear but **not** a scalar product. Which axiom fails?

2 Norm, Cauchy–Schwarz, and angle

With a scalar product in hand we can finally **measure**. The squared length of a vector is $\langle u, u \rangle$; taking a square root gives its length, or **norm**. The deeper miracle is that the scalar product also encodes **angle** — and the bridge between the two is a single inequality, the most important in the course.

2.1 The norm

Definition 3 In a Euclidean space, the **norm** (or **length**) of a vector u is

$$\|u\| = \sqrt{\langle u, u \rangle}.$$

A vector with $\|u\| = 1$ is a **unit vector**. For any $u \neq 0$, the vector $u/\|u\|$ is a unit vector in the same direction; replacing u by it is called **normalizing**.

The square root is well-defined precisely because positive definiteness makes $\langle u, u \rangle \geq 0$. In the standard product on $\mathbb{R}^n$ this is $\|u\| = \sqrt{u_1^2 + \dots + u_n^2}$, the ordinary Euclidean length. In the polynomial space it is $\|P\| = \sqrt{\int_0^1 P(x)^2 dx}$, a perfectly good “length” of a polynomial.

Proposition 4 (Basic properties of the norm) For all u and scalars λ :

- $\|u\| \geq 0$, with $\|u\| = 0$ if and only if $u = 0$;
- $\|\lambda u\| = |\lambda| \|u\|$ (absolute **homogeneity**).

2.2 The Cauchy–Schwarz inequality

Here is the keystone. It bounds the scalar product of two vectors by the product of their lengths — and it is what allows angle to be **defined**.

Theorem 5 (Cauchy–Schwarz inequality) For all vectors u, v in a Euclidean space,

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

with **equality** if and only if u and v are **collinear** (one is a scalar multiple of the other).

Proof. If $v = 0$ both sides are 0. Otherwise, for every real t consider the non-negative quantity

$$0 \leq \|u - tv\|^2 = \langle u - tv, u - tv \rangle = \|u\|^2 - 2t\langle u, v \rangle + t^2\|v\|^2.$$

This is a quadratic in t that is never negative, so its discriminant is ≤ 0 :

$$(2\langle u, v \rangle)^2 - 4\|v\|^2\|u\|^2 \leq 0 \implies \langle u, v \rangle^2 \leq \|u\|^2\|v\|^2.$$

Taking square roots gives the inequality. Equality forces the discriminant to be zero, i.e. the quadratic has a real root t_0 with $\|u - t_0 v\| = 0$, hence $u = t_0 v$ — collinearity. $\square$

Worked example — Cauchy–Schwarz in action. For $u = (1, 2, 2)$ and $v = (2, 0, -1)$ in $\mathbb{R}^3$: $\langle u, v \rangle = 2 + 0 - 2 = 0$, $\|u\| = \sqrt{9} = 3$, $\|v\| = \sqrt{5}$. Indeed $|0| = 0 \leq 3\sqrt{5}$. The vectors are far

from collinear, and the gap in the inequality is large — in fact here the scalar product is exactly 0, the case of **orthogonality** studied next.

A first dividend of Cauchy–Schwarz is that norms add the way lengths should.

Proposition 6 (Triangle inequality) $\|u + v\| \leq \|u\| + \|v\|$ for all u, v .

Proof. Expanding and applying Cauchy–Schwarz,

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2.$$

Take square roots. □

2.3 Angle and cosine similarity

Because Cauchy–Schwarz guarantees that the ratio $\langle u, v \rangle / (\|u\|\|v\|)$ always lies in $[-1, 1]$, it is exactly the cosine of an angle.

Definition 7 The **angle** $\theta \in [0, \pi]$ between two non-zero vectors u, v is defined by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

In data science this quantity is called the **cosine similarity** of u and v .

The definition is legitimate **only** because Cauchy–Schwarz keeps the right-hand side within $[-1, 1]$, the range of cosine. This is the payoff of the inequality: it is what makes “angle” meaningful in spaces — polynomials, matrices, documents-as-word-count-vectors — where no protractor could ever reach.

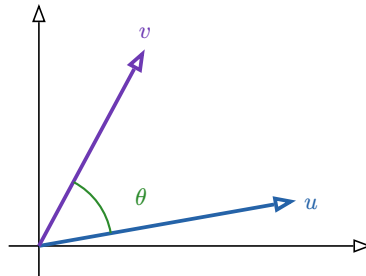


Figure 1: The angle θ between u and v is recovered from the scalar product alone: $\cos \theta = \langle u, v \rangle / (\|u\|\|v\|)$. Cauchy–Schwarz guarantees this ratio lies in $[-1, 1]$, so the angle always exists.

Parallel vectors give $\cos \theta = \pm 1$; perpendicular ones give $\cos \theta = 0$.

Common pitfall Cosine similarity measures **direction**, not magnitude. The vectors $(1, 1)$ and $(1000, 1000)$ have cosine similarity 1 — they are perfectly “similar” though hugely different in length. This is exactly why it suits comparing documents of very different sizes, but it also means a near-1 cosine says nothing about how **large** the vectors are.

Exercises

5. Compute the angle between $(1, 0)$ and $(1, 1)$ in $\mathbb{R}^2$ under the standard product.

6. Verify Cauchy–Schwarz for $u = (1, 1, 1)$, $v = (1, 2, 3)$ and determine whether equality holds.
7. Using the polynomial inner product $\langle P, Q \rangle = \int_0^1 PQ$, find the cosine of the angle between $P = 1$ and $Q = x$.
8. Prove the **parallelogram law** $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$ by expanding both sides with the scalar product.

3 Orthogonality and orthonormal bases

The single most useful angle is the right angle. When $\cos \theta = 0$ — that is, when $\langle u, v \rangle = 0$ — the vectors are **orthogonal**, and orthogonality turns out to be the organizing principle of the entire course. Families of mutually orthogonal vectors are automatically independent, and bases built from unit-length orthogonal vectors make every computation effortless.

3.1 Orthogonal and orthonormal families

Definition 8 Two vectors u, v are **orthogonal**, written $u \perp v$, if $\langle u, v \rangle = 0$. A family $(e_1, \dots, e_k)$ is

- **orthogonal** if its vectors are pairwise orthogonal: $\langle e_i, e_j \rangle = 0$ whenever $i \neq j$;
- **orthonormal** if it is orthogonal and every vector is a unit vector: $\langle e_i, e_j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$.

The standard basis $(e_1, \dots, e_n)$ of $\mathbb{R}^n$ is the prototype orthonormal family. Orthonormality is the cleanest possible relationship a family can have: it says the vectors are mutually perpendicular **and** individually of unit length.

Worked example — an orthonormal pair. In $\mathbb{R}^2$ the vectors $e_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ satisfy $\langle e_1, e_2 \rangle = \frac{1}{2}(1 - 1) = 0$ and $\|e_1\| = \|e_2\| = 1$. They are the standard axes rotated by 45° — a different orthonormal basis of the same plane.

3.2 Orthogonality forces independence

Here is the first structural payoff, and it is a strong one.

Proposition 9 (Orthogonal $\implies$ independent) Any orthogonal family of **non-zero** vectors is linearly independent.

Proof. Suppose $(e_1, \dots, e_k)$ is orthogonal with each $e_i \neq 0$, and $a_1 e_1 + \dots + a_k e_k = 0$. Take the scalar product of both sides with a fixed e_j . By bilinearity and orthogonality, every term $a_i \langle e_i, e_j \rangle$ vanishes except $i = j$, leaving

$$a_j \langle e_j, e_j \rangle = 0.$$

Since $e_j \neq 0$, positive definiteness gives $\langle e_j, e_j \rangle > 0$, forcing $a_j = 0$. As j was arbitrary, all coefficients are zero. $\square$

Remark This is genuinely useful: checking independence normally means row-reducing a matrix, but for an orthogonal family there is **nothing to do** — independence is automatic. In particular, n non-zero orthogonal vectors in an n -dimensional space form a basis for free.

Common pitfall The vectors must be **non-zero**. The zero vector is orthogonal to everything (including itself), so a family containing 0 is “orthogonal” in the literal sense yet certainly not independent. Orthonormal families dodge this automatically, since unit vectors are never zero.

3.3 Coordinates in an orthonormal basis come for free

The reason orthonormal bases are prized is that coordinates, which normally require solving a linear system, are read off by **single scalar products**.

Proposition 10 (Fourier coefficients) If $(e_1, \dots, e_n)$ is an orthonormal basis of E , then every $v \in E$ expands as

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

The coordinate of v along e_i is simply $\langle v, e_i \rangle$. Moreover (**Pythagoras / Parseval**) $\|v\|^2 = \sum_{i=1}^n \langle v, e_i \rangle^2$.

Proof. Write $v = \sum_j a_j e_j$ in the basis. Take $\langle \cdot, e_i \rangle$: orthonormality kills every term but $j = i$, giving $\langle v, e_i \rangle = a_i$. For the norm, expand $\langle v, v \rangle = \sum_{i,j} a_i a_j \langle e_i, e_j \rangle = \sum_i a_i^2$. $\square$

Worked example — reading off coordinates. With the rotated basis $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ and $v = (3, 1)$:

$$\langle v, e_1 \rangle = \frac{3+1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}, \quad \langle v, e_2 \rangle = \frac{3-1}{\sqrt{2}} = \sqrt{2}.$$

So $v = 2\sqrt{2}e_1 + \sqrt{2}e_2$, and indeed $\|v\|^2 = 8 + 2 = 10 = 3^2 + 1^2$. $\checkmark$ No system-solving was needed — just two dot products.

This single property is why the rest of the course works so hard to **build** orthonormal bases. The tool that builds them is next.

Exercises

9. Show that $(1, 2, 2)$, $(2, 1, -2)$, $(2, -2, 1)$ are pairwise orthogonal in $\mathbb{R}^3$, then normalize them to an orthonormal basis.
10. Using the orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$, find the coordinates of $(5, -1)$ and verify Pythagoras.
11. Are the polynomials 1 and $x - \frac{1}{2}$ orthogonal under $\langle P, Q \rangle = \int_0^1 PQ$? Justify.
12. Explain why any 4 pairwise-orthogonal non-zero vectors in $\mathbb{R}^4$ must be a basis, without any computation.

4 The Gram–Schmidt process

Orthonormal bases are wonderful, but a basis handed to you is rarely orthonormal. The **Gram–Schmidt process** is the algorithm that repairs this: it takes **any** basis and manufactures an orthonormal one occupying exactly the same space, one vector at a time.

4.1 The idea: subtract off what overlaps

The core move is to take a new vector and **remove its component along the directions already handled**, leaving a remainder orthogonal to all of them. The component of v along a unit vector e is $\langle v, e \rangle e$; subtracting it leaves $v - \langle v, e \rangle e$, which is orthogonal to e by construction.

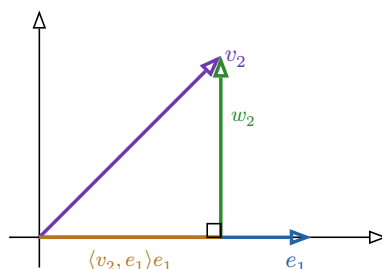


Figure 2: One Gram–Schmidt step. From v_2 subtract its component along the already-fixed direction e_1 ; the remainder $w_2 = v_2 - \langle v_2, e_1 \rangle e_1$ is orthogonal to e_1 (right angle marked). Normalizing w_2 gives e_2 . Each new vector is purged of every direction already in the basis.

4.2 The algorithm

Theorem 11 (Gram–Schmidt process) Let $(v_1, \dots, v_k)$ be a linearly independent family in a Euclidean space. Define recursively

$$w_1 = v_1, \quad w_j = v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} w_i \quad (j = 2, \dots, k),$$

and set $e_j = w_j / \|w_j\|$. Then $(e_1, \dots, e_k)$ is **orthonormal** and, for every j ,

$$\text{span}(e_1, \dots, e_j) = \text{span}(v_1, \dots, v_j).$$

The displayed **span-preservation** property is as important as the orthonormality: Gram–Schmidt does not merely produce **some** orthonormal family, it produces one that fills the **same nested subspaces** as the original. The first j outputs span exactly what the first j inputs did.

Proof. By induction each w_j is, by construction, the part of v_j orthogonal to all earlier w_i : taking $\langle \cdot, w_m \rangle$ for $m < j$ and using that the earlier w_i are already mutually orthogonal, every term cancels except the one that subtracts off the overlap, giving $\langle w_j, w_m \rangle = 0$. Since each w_j differs from v_j only by a combination of earlier vectors, the spans coincide at each stage. Normalizing rescales without changing direction or span. Independence of the v_i guarantees no w_j is zero, so the division is legitimate. $\square$

Worked example — Gram–Schmidt in $\mathbb{R}^3$. Orthonormalize $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$.

Step 1. $w_1 = (1, 1, 0)$, $\|w_1\| = \sqrt{2}$, so $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

Step 2. $\langle v_2, w_1 \rangle = 1$, $\langle w_1, w_1 \rangle = 2$, so

$$w_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right),$$

with $\|w_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$, giving $e_2 = \sqrt{\frac{2}{3}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \frac{1}{\sqrt{6}}(1, -1, 2)$.

Step 3. $\langle v_3, w_1 \rangle = 1$ and $\langle v_3, w_2 \rangle = -\frac{1}{2} + 1 = \frac{1}{2}$:

$$w_3 = (0, 1, 1) - \frac{1}{2}(1, 1, 0) - \frac{\frac{1}{2}}{\frac{3}{2}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = (0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}\right).$$

This is $w_3 = (-\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, with $\|w_3\| = \frac{2}{\sqrt{3}}$, so $e_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$. The three e_i are orthonormal — check, e.g., $\langle e_1, e_2 \rangle = \frac{1-1+0}{\sqrt{12}} = 0$. ✓

Common pitfall Two slips are common. First, when not normalizing as you go, you must divide by $\langle w_i, w_i \rangle$ — **not** by $\langle v_i, v_i \rangle$ — in the projection coefficients; project onto the **already-orthogonalized** w_i , never the raw v_i . Second, the process needs the input to be **independent**; if the v_j are dependent, some w_j comes out 0 and the normalization divides by zero — the signal that v_j was already in the span of its predecessors.

4.3 Application to function and polynomial spaces

Because Gram–Schmidt uses only the scalar product, it runs unchanged in spaces of polynomials or functions, where it produces families of **orthogonal polynomials**.

Worked example — Legendre polynomials. Apply Gram–Schmidt to $1, x, x^2$ on $\mathbb{R}[X]_{\leq 2}$ with $\langle P, Q \rangle = \int_{-1}^1 PQ \, dx$. Start with $w_1 = 1$. Then $\langle x, 1 \rangle = \int_{-1}^1 x \, dx = 0$, so x is **already** orthogonal to 1: $w_2 = x$. For x^2 , $\langle x^2, 1 \rangle = \frac{2}{3}$ and $\langle 1, 1 \rangle = 2$, while $\langle x^2, x \rangle = 0$, so

$$w_3 = x^2 - \frac{\frac{2}{3}}{2} \cdot 1 = x^2 - \frac{1}{3}.$$

The outputs $1, x, x^2 - \frac{1}{3}$ are the first **Legendre polynomials** (up to scaling) — the orthogonal basis underlying Gaussian quadrature and many approximation schemes.

Exercises

13. Apply Gram–Schmidt to $v_1 = (3, 0)$, $v_2 = (1, 2)$ in $\mathbb{R}^2$.
14. Orthonormalize $v_1 = (1, 1, 1)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 0, 0)$.
15. On $\mathbb{R}[X]_{\leq 1}$ with $\langle P, Q \rangle = \int_0^1 PQ$, orthonormalize 1 and x .
16. What happens if you run Gram–Schmidt on $v_1 = (1, 2)$, $v_2 = (2, 4)$? Interpret the result.

5 Orthogonal matrices

Gram–Schmidt produces orthonormal bases; we now study the matrices that **map one orthonormal basis to another**. These are the **orthogonal matrices** — the matrices that preserve all lengths and angles — and they are exactly the rotations and reflections of Euclidean space.

5.1 Definition and the column criterion

Definition 12 A matrix $P \in M_n(\mathbb{R})$ is **orthogonal** if

$$P^T P = I, \quad \text{equivalently} \quad P^T = P^{-1}.$$

The condition has a vivid reading in terms of columns. The (i, j) entry of $P^T P$ is the dot product of column i with column j of P ; demanding $P^T P = I$ says those dot products are 1 on the diagonal and 0 off it.

Proposition 13 (Orthogonal = orthonormal columns) P is orthogonal if and only if its columns form an **orthonormal basis** of $\mathbb{R}^n$. (Equivalently, since $PP^T = I$ too, its rows do as well.)

Worked example — a rotation matrix. The rotation by angle θ ,

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

has columns $(\cos \theta, \sin \theta)$ and $(-\sin \theta, \cos \theta)$: each has norm $\sqrt{\cos^2 + \sin^2} = 1$, and their dot product is $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$. So R is orthogonal for every θ .

5.2 Orthogonal maps preserve the geometry

The defining property is exactly preservation of the scalar product, hence of all lengths and angles.

Proposition 14 (Isometry) If P is orthogonal then for all u, v ,

$$\langle Pu, Pv \rangle = \langle u, v \rangle, \quad \text{and in particular} \quad \|Pu\| = \|u\|.$$

Proof. $\langle Pu, Pv \rangle = (Pu)^T(Pv) = u^T P^T P v = u^T I v = \langle u, v \rangle$. Setting $v = u$ gives the norm statement. $\square$

So an orthogonal matrix is a **rigid motion fixing the origin**: it can turn space or flip it, but it never stretches, shrinks, or shears. This is why orthonormal bases are interchangeable — the change-of-basis matrix between any two of them is orthogonal.

5.3 Rotations versus reflections: the determinant criterion

Taking determinants of $P^T P = I$ pins down a sharp dichotomy.

Proposition 15 If P is orthogonal then $\det P = \pm 1$.

Proof. $\det(P^T P) = \det(P^T) \det(P) = (\det P)^2$, and this equals $\det I = 1$, so $\det P = \pm 1$. $\square$

Definition 16 An orthogonal matrix with $\det P = +1$ is a **rotation** (it preserves orientation); one with $\det P = -1$ is a **reflection** / orientation-reversing (it flips orientation, like a mirror).

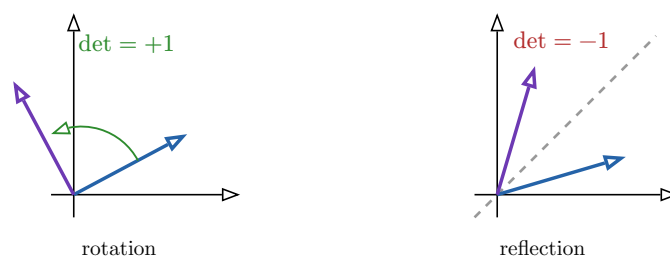


Figure 3: Both kinds of orthogonal map preserve lengths and angles. A **rotation** ($\det = +1$, left) keeps orientation: the ordered pair (blue, purple) stays counter-clockwise. A **reflection** ($\det = -1$, right) flips a vector across a mirror line (dashed), reversing orientation.

Common pitfall $\det P = -1$ does **not** by itself make P orthogonal — orthogonality is the condition $P^T P = I$, and the determinant test only **classifies** a matrix already known to be orthogonal. Check $P^T P = I$ first; then the sign of the determinant tells you rotation versus reflection.

Remark Orthogonal matrices form a group: products and inverses of orthogonal matrices are orthogonal (if $P^T P = I$ and $Q^T Q = I$ then $(PQ)^T (PQ) = Q^T P^T P Q = I$). Composing two rotations gives a rotation; composing two reflections gives a rotation $((-1)(-1) = +1)$ — two mirrors make a turn.

Exercises

17. Verify that $\frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}$ is orthogonal and decide whether it is a rotation or a reflection.
18. Show the reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is orthogonal with determinant -1 , and describe geometrically what it does.
19. Prove that if P and Q are orthogonal then so is PQ .
20. For which θ is the rotation $R(\theta)$ equal to its own inverse? Interpret the answer.

6 Symmetric matrices and the spectral theorem

We now bring the geometry of orthogonality to bear on the eigenvalue theory of MAT21-1. The result is the crown jewel of the course. For a special but ubiquitous class of matrices — the **symmetric** ones — diagonalization always succeeds, the eigenvalues are always real, and the eigenvectors can always be chosen mutually **orthogonal**. None of the pathologies of MAT21-1 (complex roots, eigenvector shortfalls) can occur.

6.1 Symmetric matrices and self-adjointness

Definition 17 A matrix $A \in M_n(\mathbb{R})$ is **symmetric** if $A^T = A$, i.e. $a_{ij} = a_{ji}$.

The reason symmetry interacts so well with the scalar product is the following identity, which says A can be moved from one side of an inner product to the other.

Proposition 18 (Self-adjointness) A is symmetric if and only if $\langle Au, v \rangle = \langle u, Av \rangle$ for all $u, v \in \mathbb{R}^n$.

Proof. $\langle Au, v \rangle = (Au)^T v = u^T A^T v$ and $\langle u, Av \rangle = u^T Av$. These agree for all u, v exactly when $A^T = A$. $\square$

6.2 Two structural facts

Symmetry forces two remarkable properties before we even reach diagonalization.

Proposition 19 (Real eigenvalues) Every eigenvalue of a real symmetric matrix is **real**.

Proof. Let $Av = \lambda v$ with $v \neq 0$, allowing complex λ, v for the argument, and let $\bar{v}$ be the entrywise conjugate. Then $\bar{v}^T Av = \lambda \bar{v}^T v$. Taking conjugate-transpose of this scalar and using $A^T = A$ with real entries gives $\bar{v}^T Av = \bar{\lambda} \bar{v}^T v$. Since $\bar{v}^T v = \sum |v_i|^2 > 0$, we get $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$. $\square$

Proposition 20 (Orthogonal eigenspaces) Eigenvectors of a symmetric matrix for **distinct** eigenvalues are orthogonal.

Proof. Let $Au = \lambda u$ and $Av = \mu v$ with $\lambda \neq \mu$. Using self-adjointness,

$$\lambda \langle u, v \rangle = \langle Au, v \rangle = \langle u, Av \rangle = \mu \langle u, v \rangle,$$

so $(\lambda - \mu) \langle u, v \rangle = 0$. As $\lambda \neq \mu$, we conclude $\langle u, v \rangle = 0$. $\square$

These two facts are precisely what is needed: real eigenvalues mean the characteristic polynomial splits over $\mathbb{R}$, and orthogonal eigenspaces mean the eigenvectors assemble into an **orthonormal** basis. The spectral theorem packages this.

6.3 The spectral theorem

Theorem 21 (Spectral theorem) Every real symmetric matrix $A \in M_n(\mathbb{R})$ is **orthogonally diagonalizable**: there exist an **orthogonal** matrix P (so $P^T = P^{-1}$) and a real diagonal matrix D with

$$A = PDP^T.$$

The columns of P form an orthonormal basis of eigenvectors of A , and the diagonal of D holds the corresponding (real) eigenvalues. Conversely, any matrix of the form PDP^T with P orthogonal and D diagonal is symmetric.

The converse is the easy half and worth seeing: if $A = PDP^T$ then $A^T = (PDP^T)^T = PD^T P^T = PDP^T = A$, since D is diagonal. So **orthogonal diagonalizability and symmetry are the same property** — symmetric matrices are exactly those a rotation/reflection turns into pure scalings.

Common pitfall Note the P^T in place of the P^{-1} of ordinary diagonalization. Because P is orthogonal the two coincide, so $A = PDP^T$ — but the point is that for a symmetric matrix you may **choose** P orthogonal, which is a far stronger and cleaner statement than mere diagonalizability. Inverting P is then free: just transpose.

6.4 The recipe, with a worked case

To orthogonally diagonalize a symmetric A :

1. Find the eigenvalues via $\chi_{A(\lambda)} = \det(A - \lambda I) = 0$ (all real).
2. For each eigenvalue, find a basis of its eigenspace. Eigenvectors from **distinct** eigenvalues are already orthogonal; within a **repeated** eigenvalue's eigenspace, run **Gram-Schmidt** to orthonormalize.
3. Normalize all eigenvectors and place them as the columns of P ; place the matching eigenvalues on the diagonal of D . Then $A = PDP^T$.

Worked example — a full orthogonal diagonalization. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (symmetric). From MAT21-1, the eigenvalues are 3 and 1 with eigenvectors $(1, 1)$ and $(1, -1)$ — which are **orthogonal**, as the theory promised. Normalize:

$$P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Here P is orthogonal ($P^T P = I$), and one checks $A = P D P^T$. Compare with MAT21-1, where the same matrix needed an explicit P^{-1} ; now inversion is just transposition.

Worked example — a repeated eigenvalue needing Gram–Schmidt. Let $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, symmetric. Its eigenvalues are $\lambda = 2$ (from the top block and again from the 2×2 block, which has eigenvalues 2 and 0) and $\lambda = 0$. The eigenspace E_2 is **two-** dimensional, spanned by $(1, 0, 0)$ and $(0, 1, 1)$ — already orthogonal here, so normalize to $(1, 0, 0)$ and $\frac{1}{\sqrt{2}}(0, 1, 1)$. The eigenvector for 0 is $(0, 1, -1)$, orthogonal to both (distinct eigenvalue). Had the two vectors spanning E_2 **not** been orthogonal, Gram–Schmidt inside E_2 would have fixed that. The spectral theorem guarantees this always works.

Exercises

21. Orthogonally diagonalize $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.
22. Show directly that the eigenvectors of $\begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix}$ for its two eigenvalues are orthogonal.
23. Prove that if $A = P D P^T$ with P orthogonal and D diagonal then A is symmetric.
24. Orthogonally diagonalize $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, using Gram–Schmidt on the repeated eigenspace if needed.

7 Quadratic forms and positive definiteness

The spectral theorem has an immediate, high-value application: classifying **quadratic forms**. A symmetric matrix defines a function $q(x) = x^T A x$, and the **signs of its eigenvalues** decide whether that function is a bowl, a dome, or a saddle. This is exactly the test used in optimization to classify critical points via the **Hessian**.

7.1 Quadratic forms

Definition 22 The **quadratic form** associated with a symmetric matrix $A \in M_{n(\mathbb{R})}$ is the function

$$q(x) = x^T A x = \sum_{i,j} a_{ij} x_i x_j.$$

Worked example — reading a form off its matrix. $q(x, y) = 2x^2 + 6xy + 5y^2$ corresponds to $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$: the diagonal entries are the coefficients of x^2, y^2 , and each **off-diagonal** entry is **half** the coefficient of the cross term xy (split symmetrically as $3 + 3 = 6$). Always halve the cross terms when building A — this is the single most common error.

7.2 Classification by eigenvalue signs

Diagonalizing $A = PDP^T$ and substituting $y = P^T x$ turns the form into a pure sum of squares $q = \sum \lambda_i y_i^2$ with **no cross terms** — the **principal axes**. The sign of each λ_i then determines everything.

Definition 23 A symmetric matrix A (and its quadratic form) is

- **positive definite** if $x^T A x > 0$ for all $x \neq 0$;
- **positive semi-definite** if $x^T A x \geq 0$ for all x ;
- **negative definite** / **negative semi-definite**: the reverse inequalities;
- **indefinite** if $x^T A x$ takes both positive and negative values.

Theorem 24 (Eigenvalue classification) Let A be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$. Then A is

- **positive definite** $\iff$ all $\lambda_i > 0$;
- **positive semi-definite** $\iff$ all $\lambda_i \geq 0$;
- **negative definite** $\iff$ all $\lambda_i < 0$;
- **indefinite** $\iff$ there are eigenvalues of **both** signs.

Proof. Write $q(x) = x^T A x = (P^T x)^T D (P^T x) = \sum_i \lambda_i y_i^2$ where $y = P^T x$. As x ranges over all non-zero vectors, so does y (since P is invertible). The sign behaviour of $\sum \lambda_i y_i^2$ is then dictated entirely by the signs of the λ_i : all positive makes it strictly positive, mixed signs let it go both ways (choose y along the relevant axis), and a zero eigenvalue allows $q = 0$ on a non-zero vector (semi-definite). $\square$

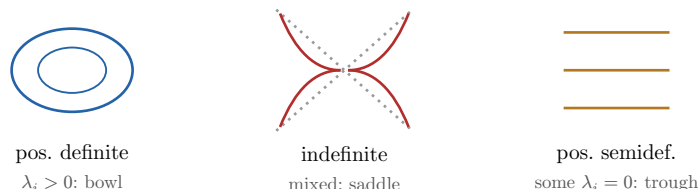


Figure 4: Level sets of a 2-variable quadratic form. Positive-definite (all $\lambda_i > 0$): nested ellipses around a minimum — a bowl. Indefinite (mixed signs): hyperbolas — a saddle, going up one way and down another. Positive semi-definite (a zero eigenvalue): a flat trough with a direction of no curvature.

7.3 The connection to the Hessian

Remark In multivariable calculus, the second-order behaviour of a function f at a critical point is captured by its **Hessian matrix** H of second partial derivatives, which is **symmetric**. The classification above is exactly the **second-derivative test**: H positive definite $\implies$ local **minimum** (the bowl); negative definite $\implies$ local **maximum** (the dome); indefinite $\implies$ a **saddle** point (neither). The eigenvalue signs of the Hessian are how optimization algorithms tell minima from saddles.

Common pitfall Reading definiteness off the diagonal entries alone is wrong. The matrix $\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ has positive diagonal yet eigenvalues 4 and -2 — it is **indefinite**. Positive diagonal

entries are necessary but nowhere near sufficient; you must check the **eigenvalues** (or use a determinant-based criterion).

Exercises

25. Write the matrix of $q(x, y) = 3x^2 - 4xy + 3y^2$ and classify it.
26. Classify $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ by their eigenvalues.
27. For which values of c is $\begin{pmatrix} 1 & c \\ c & 4 \end{pmatrix}$ positive definite?
28. Explain why a positive-definite matrix is necessarily invertible.

8 Orthogonal complement and projection

We turn to the second great theme: **approximation**. Given a subspace F and a vector v that may not lie in it, what is the point of F **closest** to v ? The answer rests on splitting any vector into a part inside F and a part exactly perpendicular to F — the **orthogonal complement** and the **orthogonal projection**.

8.1 The orthogonal complement

Definition 25 Let F be a subspace of a Euclidean space E . Its **orthogonal complement** is

$$F^\perp = \{v \in E : \langle v, f \rangle = 0 \text{ for all } f \in F\},$$

the set of vectors orthogonal to **everything** in F . It is itself a subspace.

In practice F is given by a spanning set, and a vector lies in $F^\perp$ exactly when it is orthogonal to each spanning vector — a system of linear equations.

Worked example — computing an orthogonal complement. In $\mathbb{R}^3$ let $F = \text{span}((1, 1, 0))$. A vector $(x, y, z) \in F^\perp$ iff $\langle (x, y, z), (1, 1, 0) \rangle = x + y = 0$. So $F^\perp = \{(x, -x, z)\} = \text{span}((1, -1, 0), (0, 0, 1))$, a plane — the plane of all vectors perpendicular to the line F .

8.2 The direct sum decomposition

The central structural fact is that F and $F^\perp$ together fill all of E , with no overlap.

Theorem 26 (Orthogonal decomposition) For any subspace F of a finite-dimensional Euclidean space E ,

$$E = F \oplus F^\perp, \quad \text{hence} \quad \dim F + \dim F^\perp = \dim E.$$

Every $v \in E$ has a **unique** decomposition $v = v_F + v_\perp$ with $v_F \in F$ and $v_\perp \in F^\perp$.

Proof. **No overlap:** if $w \in F \cap F^\perp$ then w is orthogonal to itself, $\langle w, w \rangle = 0$, so $w = 0$ by positive definiteness. **Spanning:** take an orthonormal basis $(e_1, \dots, e_k)$ of F (Gram–Schmidt guarantees one), and for any v set $v_F = \sum_i \langle v, e_i \rangle e_i \in F$. Then $v - v_F$ is orthogonal to every e_i , hence to all of F , so it lies in $F^\perp$. Thus $v = v_F + (v - v_F)$ is the required decomposition; the dimension formula follows from the direct sum. $\square$

Common pitfall The dimension formula $\dim F + \dim F^\perp = \dim E$ is the tool to **find** $\dim F^\perp$ without computing it directly: a plane in $\mathbb{R}^3$ ($\dim F = 2$) has a **line** as complement ($\dim F^\perp = 1$), and vice versa. But it depends on the ambient E : the complement of a line is a 4-dimensional space inside $\mathbb{R}^5$, not the same as in $\mathbb{R}^3$. The complement is always taken **inside a stated ambient space**.

8.3 Orthogonal projection

The “ F -part” v_F of the decomposition is the orthogonal projection.

Definition 27 The **orthogonal projection** of v onto F is the unique $p_F(v) = v_F \in F$ from the decomposition above. If $(e_1, \dots, e_k)$ is an **orthonormal** basis of F , then

$$p_F(v) = \sum_{i=1}^k \langle v, e_i \rangle e_i.$$

Common pitfall The formula $p_{F(v)} = \sum \langle v, e_i \rangle e_i$ requires the basis to be **orthonormal**. With a merely orthogonal basis, divide each term by $\langle e_i, e_i \rangle$; with a general (non-orthogonal) basis the formula is simply wrong — orthonormalize first via Gram-Schmidt, or use the projection matrix below.

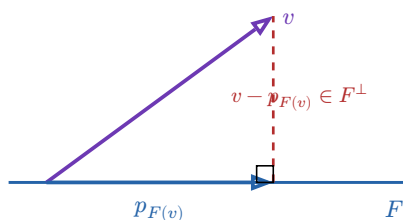


Figure 5: Orthogonal projection onto a subspace F . The vector v splits into $p_{F(v)} \in F$ (its shadow on F) and the perpendicular remainder $v - p_{F(v)} \in F^\perp$ (the right angle is marked). This is the unique way to write v as “something in F plus something perpendicular to F ”.

8.4 The projection matrix

When F is a subspace of $\mathbb{R}^n$, the projection is a linear map, hence a matrix.

Proposition 28 (Projection matrix) If the columns of Q form an **orthonormal** basis of $F \subset \mathbb{R}^n$, the orthogonal projection onto F is left-multiplication by

$$P_F = QQ^T.$$

More generally, if the columns of M are **any** basis of F (not necessarily orthonormal), then

$$P_F = M(M^T M)^{-1} M^T.$$

In both cases P_F is symmetric and **idempotent** ($P_F^2 = P_F$): projecting twice is the same as projecting once.

Worked example — projecting onto a line. Project $v = (3, 4)$ onto the line $F = \text{span}((1, 0))$ in $\mathbb{R}^2$. The unit vector is $e = (1, 0)$, so $p_{F(v)} = \langle v, e \rangle e = 3(1, 0) = (3, 0)$ — drop

the second coordinate. The projection matrix is $P_F = ee^T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, and indeed $P_F(3, 4)^T = (3, 0)^T$. The remainder $(0, 4) \in F^\perp$.

Exercises

29. Find $F^\perp$ for $F = \text{span}((1, 2, 2))$ in $\mathbb{R}^3$ and check the dimension formula.
30. Project $v = (1, 2, 3)$ onto $F = \text{span}((1, 0, 0), (0, 1, 0))$ and identify the perpendicular remainder.
31. Build the projection matrix onto $\text{span}((1, 1))$ in $\mathbb{R}^2$ and verify it is symmetric and idempotent.
32. Show that $I - P_F$ is the projection onto $F^\perp$.

9 Best approximation, distance, and least squares

Everything assembled now yields the theorem the course was built for: the orthogonal projection of v onto F is the **closest point of F to v** . This is the **best-approximation theorem**, and its distance is the residual that least-squares regression minimizes.

9.1 The best-approximation theorem

Theorem 29 (Best approximation) Let F be a subspace of a Euclidean space and v any vector. The orthogonal projection $p_{F(v)}$ is the **unique closest point of F to v** : for every $f \in F$ with $f \neq p_{F(v)}$,

$$\|v - p_{F(v)}\| < \|v - f\|.$$

Proof. For any $f \in F$, write $v - f = (v - p_{F(v)}) + (p_{F(v)} - f)$. The first bracket lies in $F^\perp$ and the second in F , so they are orthogonal, and Pythagoras gives

$$\|v - f\|^2 = \|v - p_{F(v)}\|^2 + \|p_{F(v)} - f\|^2.$$

The second term is ≥ 0 , and is zero only when $f = p_{F(v)}$. Hence $\|v - f\|$ is strictly larger than $\|v - p_{F(v)}\|$ unless $f = p_{F(v)}$. $\square$

The proof is the whole idea in one line: **the error $v - p_{F(v)}$ is perpendicular to F , and a perpendicular dropped to a subspace is shorter than any slanted line to it.**

9.2 Distance to a subspace

Definition 30 The **distance** from v to a subspace F is the minimal distance to its points, achieved at the projection:

$$d(v, F) = \|v - p_{F(v)}\| = \|v_\perp\|.$$

Worked example — distance from a point to a plane. Find the distance from $v = (1, 1, 1)$ to the plane $F = \text{span}((1, 0, 0), (0, 1, 0))$ (the xy -plane) in $\mathbb{R}^3$. The projection drops the third coordinate: $p_{F(v)} = (1, 1, 0)$. The remainder is $v - p_{F(v)} = (0, 0, 1) \in F^\perp$, so

$$d(v, F) = \|(0, 0, 1)\| = 1.$$

The closest point of the plane is $(1, 1, 0)$, directly “below” v .

9.3 Why this is least-squares regression

Remark Least-squares regression fits a model to data by minimizing the sum of squared errors — and that is **exactly** a distance minimization in a Euclidean space. To solve an overdetermined system $Ax \approx b$ (more equations than unknowns, no exact solution), we seek the x making Ax as close to b as possible. The achievable vectors Ax form the column space $F = \text{Im}(A)$, so the best Ax is the **orthogonal projection of b onto F** . The error $b - Ax$ must be perpendicular to F , i.e. $A^T(b - Ax) = 0$, which rearranges to the **normal equations**

$$A^T A x = A^T b.$$

Least-squares regression **is** the best-approximation theorem, and the regression line is the orthogonal projection of the data onto the space of fitted values.

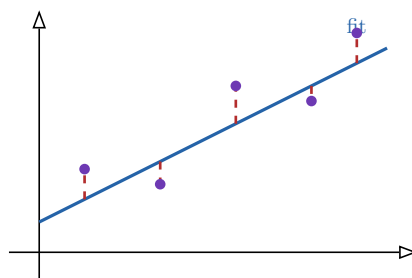


Figure 6: Least-squares regression as best approximation. The dashed residuals are the gaps between data (purple) and the fitted line (blue). Minimizing the sum of their squares is exactly projecting the data vector b orthogonally onto the column space of the design matrix — the closest reachable point, with error perpendicular to that space.

The arc in one breath. A **scalar product** equips a vector space with length ($\|u\| = \sqrt{\langle u, u \rangle}$) and, through **Cauchy–Schwarz**, angle ($\cos \theta = \langle u, v \rangle / (\|u\| \|v\|)$ — cosine similarity). **Orthogonal** families are automatically independent, and **orthonormal** bases make coordinates free; **Gram–Schmidt** builds them from any basis. **Orthogonal matrices** ($P^T = P^{-1}$) are the rigid motions — rotations ($\det = +1$) and reflections ($\det = -1$). For **symmetric** matrices the **spectral theorem** guarantees an orthonormal eigenbasis, $A = PDP^T$; the **signs of the eigenvalues** classify the quadratic form and, via the Hessian, the critical points of optimization. Finally, splitting $E = F \oplus F^\perp$ gives the **orthogonal projection**, whose defining virtue — the **best-approximation theorem** — is that it is the closest point of F , the perpendicular error being shorter than any other. That distance is the residual of **least-squares regression**. The scalar product turned algebra into geometry, and geometry answered “how close?”.